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CODING OF SNG for STEP 2

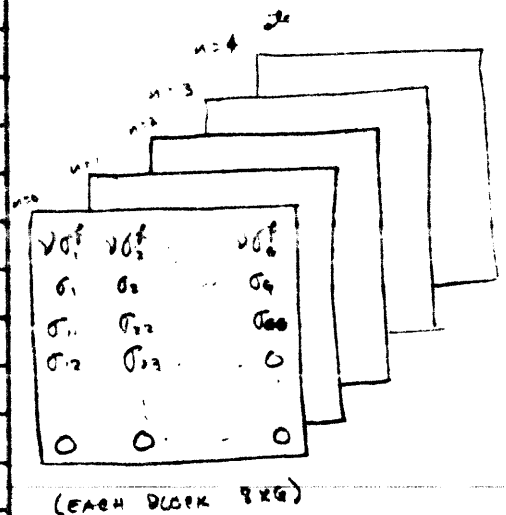
SNG: X-Sec, Mm.

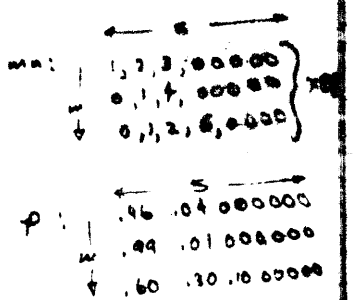
SEQ	OPN OP	SAUCE	A OA	GAB	X X	XB	SA SA	SAB	COMMENTS
K	TS		K+1		1		R_0		
K+1	TS		K+2		2		R_1		
2	TS		K+3		3		R_2		
3	L	S	L(0)				Σ		
4	L	S	L(m)	1	(1,3)	G	4		
5	L		L(P_0)	1	(1,3)	G			
6	BB		K+10				#1		
7	M		L(σ_0)	1	(2,4)	G			
8	A	S	Σ				Σ		
9	IBR		K+4		3		1		
10	ST		L(σ_0)	1	(2,4)	G			
11	TR		K+12		3				
12	IBR		K+3		2		1		
13	IB1		K+3		1		8		
14	BB		(TO MONITOR PRINT)				#4		
15	TS		K+25		1		C(0,I,R)		
16									
17									
18									
19									
20									
21									
22									
23									
24									

m					
m	R_3 → 1	0	M	—	M-1
t	R_1 → 2	0	84	R_1	A=2
s	R_2 → 3	0	8	R_2	A=1
m	→ 4	← EXT →			

NOTES: 1. $\sigma_{t,m} = \sum_s p_{s,m}\,\sigma_{s,s,m}$

2. WE ASSUME STORAGES AS INDICATED BELOW.





SNG: GEOMETRY

SEQ	OPN	A	X	SA	COMMENTS
α+25	A L	(r_0) 1	(1,2) G	$(\bar{r}_0)$ 1	(1)
26	ES S	/		$(\bar{r}_0)$ 1	(1)
27	S S	(r_0) 1	(1,2) G	(Δ_0) 1	(1)
28	D S	$(\bar{r}_0)$ 1	(1)	(s_0) 1	(1)
29	L	(Δ_0) 1	(1).		
30	TSL	α+31	L(T)		
31	M	(r_0) 1	(1)		
32	M	(r_0) 1	(1)		
33	ST	(t_0) 1	(1)		
34	IBR	α	1	1	
35					
36					
37					
38					
39					
40					
41					
42					
43					
44					
45					
46					
47					
48					
49					

i:	1	0	I	R_0
-1:	2	-1	—	R_α
T	L(T)	0	—	—

NOTES: 1. $\bar{r}_i = \frac{r_i + r_{i-1}}{2}$

2. $\Delta_i = \frac{r_i - r_{i-1}}{2}$

3. $s_i = \Delta_i / \bar{r}_i$

4. $t_i = \Delta_i$ (P)
$= \Delta r_i$ (O)
$= \Delta \bar{r}_i$ (S)

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SUB: INITIALIZE R4 & LOOP; MEP: Control

SEQ	OPN	A	X	SA	COMMENTS
(α+ 50	TS	α+1	3	R_5	
51	TS	α+2	4	R_6	
52	TS	α+3	5	R_7	
53	L	L(0)			
54	BNP	α+66		#1	*
55	D L	(V_g) 1	(3)	α	
56	ST	T_0			
57	L S	(a_{-1}) 1	(1)	6	
58	M L	$(G_0 4)$ 1	(3, 6) G	(P_0) 1	(1)
59	A	T_0			
60	M	(λ_0) 1	(1)		
61	BB	α+63		#2	
62	L	L(0)			
63	ST	(ℓ_0) 1	(1)		
64	L S	L(0)		Σ	
65	L	L(IT σ_T)			
66	BB	α+75		#1	
01	L	(σ_{0+24}) 1	(3,5,6) G		
68	BB	α+73		#1	
69	M	(N_{g+1}) 1	(1,4) G		
70	A S	Σ		Σ	
71	ICBR	α+72	4	R_3	
72	ICBR	α+67	5	R_5	
73	M L	(ρ_0) 1	1	Σ	
74	ST	Σ			

i	1	0	I	R_3
-1	2	-1	—	R_4
g	3	G	G	R_5
gI	4	(G-1)I	1	R_6
gG	5	0	1	R_7
M	6	← SET		→

NOTES: 1. $\rho(\tau_0) = \alpha/V_g$ or 0.

2. $\ell_i = \Delta_i \left[\sigma_{g+i} \cdot \rho_i + \alpha/V_g\right]$

3. $\delta\ell_i = 2\Delta_i \left[V_g \ell_i + \rho_i \sum_{g}^{g+4} N_{g+i} \cdot \sigma_{g+i}\right]$

4. ρ #1 ON = $\underline{\alpha}$ calc.
   OFF = $\underline{\alpha}$ calc

5. IF $\ell_i < 0$, $0 \longrightarrow \ell_i$

6. IF $IT \cdot \sigma_T = 0$, $0 \longrightarrow \Sigma$ for δ_i.

(3)

SEQ	OPN	A	X	SA	COMMENTS
α+75	M L	(V_q) 1	(3)	(f_i) 1	(1)
76	A	Σ			
77	M	(Δ_i) 1	(1)		
78	AE S	1		(β_i) 1	(1)
79	JR	α+80	4,5 G		
80	IBR	α+37	1	1	
81	S L	(N_0) 1	(L(I-1))		→ B.C $N_I(j) = 0$, $j = 0,1,2$
82	S L	(N_0) 1	(L(32-1))		
83	S L	(N_0) 1	(L(32-1))		
84	L S	L(0)		7	→ [illegible] $\frac{1}{2}$ [illegible] y_{pk} = 0 if $p = y_{pk}$
85	BNP	α+87			
86	TS	α+87	7	L(0, x_2, R)	
87					
88					
89					
90					
91					
92					
93					
94					
95					
96					
97					
98					
99					

SNG: Calc $N_{i,j}$

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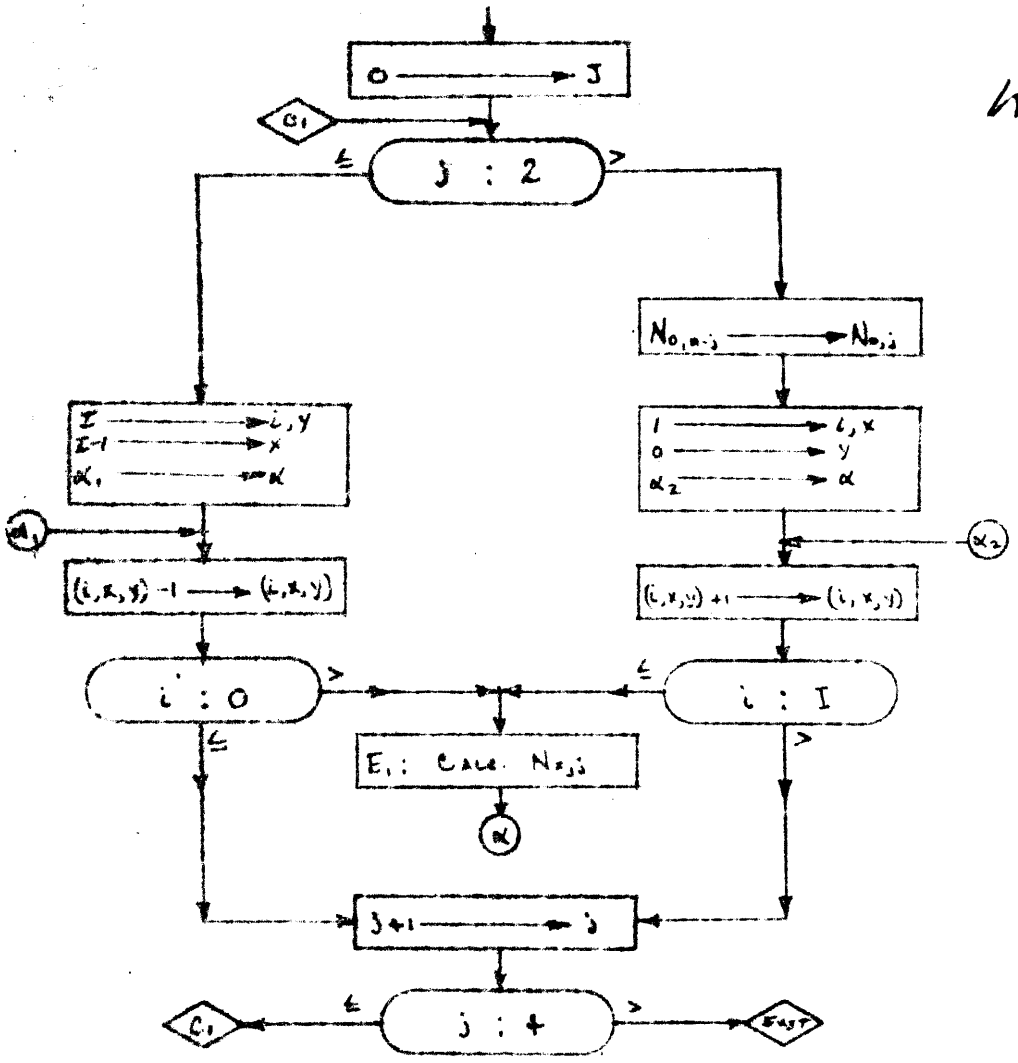
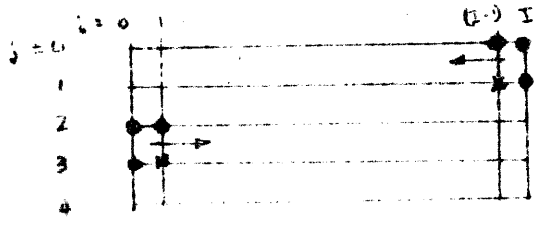


$$E1: \quad N_{x,j} = \frac{(m_j - b_j s_i - k_i) N_{x,j} + (m_j + b_j s_i - k_i) N_{x,j-1} - (m_j - b_j s_i + k_i) N_{x,j-1} + \delta_i}{(m_j + b_j s_i + k_i)}$$

INWARD: $x = i-1$, $y = i$

OUTWARD: $x = i$, $y = i-1$



$N_{i,j}$ STORAGE

SNG: ANGULAR FLUX CONTROL CODE

SEQ	OPN	A	X	SA	COMMENTS
α+100	ST L	T_0	X,6	(N0) 1	(X,6)
101	BVNE	α+103	L(I)	1	
102	M S	(Σ) 1	(7)	(T_0) 1	(X,6)
103	IBR	α+100	15	1	
104	TS	α+105	4,5 G	R_{10}:(0,J,R)	
105	BRG	α+112	4	2	
106	TS	α+107	1,9,10 G	R''' L(I,I,R)	
107	TD	α+108	9	1	
108	TSL	α+110		α_R	
109	TD	α+110	1,9,10 G	1	← β_1
110	BXE	α+120	1	0	
111	TSL	α+125		—	→ β
112	L S	(N0) 1		(N0) 1	(L(4(I+1)))
113	L S	(N0) 1	L(I)	(N0) 1	(L(3(I+1)))
114	TS	α+115	1,9,10 G	R_2 : (0,I,R_2)	
115	TI	α+116	1,9 G	1	
116	TSL	α+118		α_R	
117	IBR	α+118	1,9,10 G	1	← β_2
118	BXE	α+120	1	0	
119	TSL	α+125		—	→ β
120	IBR	α+121	5	L(I+1) 1	
121	IBR	α+105	4	1	
122	BP	Point (i_1, $N_{i,j}$)		*3	*
123					
124					

i:	1	0	I	R_3
-1:	2	-1	—	—
g:	3	6	G	R_5
g I:	4	6-ΔI	I	R_{10}
g G:	5	0	1	R_9
(15):	15	0	15	R_3
k:	7	0	2	R_2
j:	4	0	J	R_{10}
j(I+1):	5	0	J	R_{10}
x:	9	Σ_0 : I_0	I	R_0
y:	10	I_1 : σ_0	I	R_{11}
-I:	11	-(I+1)	—	—

NOTES: 1. WE ASSUME $N_{i,j}$, $\bar{N}_{i,j}$, & $\sigma_{i,j}$ TO BE LOCATED HERE.

2. 11 CAN BE SET IN THE CODE INITIALIZATION.

Special Case: $N_{k2} = \dfrac{(m_j - b_j s_i - t_i) N_{kj} + s_i n_k}{(m_j + b_j s_i + t_i)}$

SAVE: N_{ki} (SUBROUTINE)

SEQ	OPN	A	X	SA	COMMENTS
0+125	M L	(b_j) ,	(4)	(s_i) L	(1) T_0: $(b_j s_i)$
26	ST	T_0			T_1: $(b_j s_i + t_i)$; $b_j s_i - t_i$
27	A S	(t_i) ,	(1)	T_1	T_2: $(m_j + b_j s_i + t_i)$ = DEN.
28	A S	(m_j) ,	(4)	T_2	T_3: Partial Num.
29	S L	T_1		(m_j) ,	(4)
30	M	(N_6) ,	(5,10) G		
31	A S	(N_4) ,	(1)	T_3	
32	BYE	0+102	4	0	
33	S L	(s_i) ,	(1)	T_0	
34	ST	T_1			
35	A	(n_i) ,	(4)		
36	M	(N_6) ,	(5,10,11) G		
37	A S	T_3		T_2	
38	S L	(m_j) ,	(4)	T_1	
39	M	(N_6) ,	(5,9,11) G		
40	A	(a_{ij}) ,	(1)		
41	A	T_3			
42	D S	T_2		(N_6) ,	(5, 9)
43	TSL	1	0/R	—	
44					
45					
46					
47					
48					
49					

SEQ	OPN	A	X	SA	COMMENTS
α+150	TI	α+151	1	1	
51	TS	α+152	4	R_0	
52	TS	α+153	5	R_5	
53	L S	L(6)		Σ	
54	A C	(N_0) 1	(1,2,4) G	(N_0) 1	(1,4) G
55	BRS	α+157	4	5	
56	MCT	α+158	4	0	
57	SF	1			
58	A S	Σ		Σ	
59	IBR	α+154	4	L(+1) 1	
60	D	L(2,0)			
61	BNP	α+165		#2	
62	M	(ω_0) 1	(7)		
63	DRE	α+165	7	2	
64	A	(N_0) 1	(1,2,5) G		
65	ST	(N_{gi}) 1	(1,2,5) G		
66	IBR	α+153	1	1	
67					
68					
69					
70					
71					
72					
73					
74					

	C	I	I	R
(I=)	4	0	I	R_0
δ^I	5	(G-1)I	–	R_5
-1	2	-1	–	–
n	7	0	Σ	R

NOTES:

1. $N_{gi} = \frac{1}{2\pi} \sum_j \delta_j [N_{cos} + N_{-sin}]$

   where $\delta_j = 1/2$, $j = 0, n$

   $\delta_j = 1$, $j \neq 0, n$

2. DP 2 on: ω_0
   on: P on ω_0

SEQ	OPN	A	X	SA	COMMENTS

SUB: I' E' L, E, D E

SEQ	OPN	A	X	SA	COMMENTS
X+200	TSL	X+225		A'R	
1	STL S	E'		I'	
2	TS	X+203	5	L(0,G,R)	
3	L S	L(0)		Σ1	
4	ST	Σ2			
5	L S	(n) 1	(1)	4	
6	M L	(A) 1	(4,5) G	(Wgi) 1	(1,3) G
7	A S	Σ1		Σ1	
8	M L	(D+G) 1	(4,5) G	(Y'gi) 1	(1,3) G
9	A S	Σ2		Σ2	
10	IBR	X+211	2	L(I) 1	
11	IBR	X+206	5	1	
12	M S	(D) 1	(1)	(E) 1	(1)
13	M L	(D) 1	(1)	Σ1	
14	ST	(D) 1	(1)		
15	IBR	X+203	1	1	
16	TSL	X+225		A'R	
17	STL S	D		D	
18					
19					
20					
21					
22					
23					
X+224					

L	1	0	I	R3
g I	3	0	G	R
n	4		3=T	
g	5	0	G	R

$\bar{E}$ $\bar{E}$ (SUBROUTINE)

SEQ	OPN	A	X	SA	COMMENTS
0+235	L S	L(0)		Σ1	
26	ST	Σ2			
27	M L	(f1)	(1)	(Σ1) 1	(1)
28	A S	Σ2		Σ2	
29	M L	(f2)	(1)	(Σ2) 1	(1)
30	A S	Σ3		Σ3	
31	TBR	0+27	1	1	
32	LL L	Σ3		Σ1	
33	TSL	1	0		
34					
35					
36					
37					
38					
39					
40					
41					
42					
43					
44					
45					
46					
47					
48					
99					

SEQ	OPN	A	X	SA	COMMENTS
50	L S	L(0)		$T_0 = \Sigma_L$	
51	M L	(4) 1	(1,3) 4	(4) 1	1)
52	D	(4) 1	(5)		
53	A S	Σ_L		Σ_L	
54	-BR	$\alpha + 258$	1	1	
55	-BR	$\alpha + 256$	3	L(I) 1	
56	-BR	$\alpha + 251$	5	1	
57	ST	$\bar{n}$			
58	L S	L(0)		Σ_L	
59	L	(5) 1	(5)		
60	TSL	$\alpha + 261$	L(T)		
61	M	(F) 1	L(I-1)		
62	M	(T) 1	L(I-1)		
63	A S	Σ_L		Σ_L	
64	-BR	$\alpha + 259$	5	1	
65	ST	ℓ			
66	M L	$\bar{e}'$		$\bar{I}'$	
67	D S	$\bar{I}'$		$\bar{I}_0'$	
68	A L	L(1)		(I-1)	
69	ST	L(I, C)			
70	S	L(3)			
71	BB	$\nu + 300$		$\nu + 3$	
72					
73					
79					

L	1	0	I	R
T	3	C	4	ℓ
g	5	0	4	R

NOTES:

1. $\bar{n} = \sum_{i,g} i_g N_{gi} / N_g$

2. $\ell = \sum_g \ell_g \cdot 1$ (P)
   $\cdot \bar{r}_I$ (C)
   $\cdot \bar{r}_I^2$ (S)

SEQ	OPN	A	X	SA	COMMENTS

Notes:

1. Pre-
2. For
3. For

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SN? ALGORITHM

SEQ	OPN	A	X	SA	COMMENTS
A + 200	TS	0+1301	4	4(2,3,4)	
1	L S	(0) 1	(2,3) 6	(0,2) 1	(4)
2	SRC	0+1301	2	1	
3	L S	(1,2) 1	( )	(0,2) 1	
4	S L	(0,2) 1	(0,1) 2	(1) 1	(5)
5	MP				
6	D	L(6)			
7	SS	A+14		5+2	
8	JBE	A+3+4	2	1	
9	TSL	(0+2 → 1+2)			